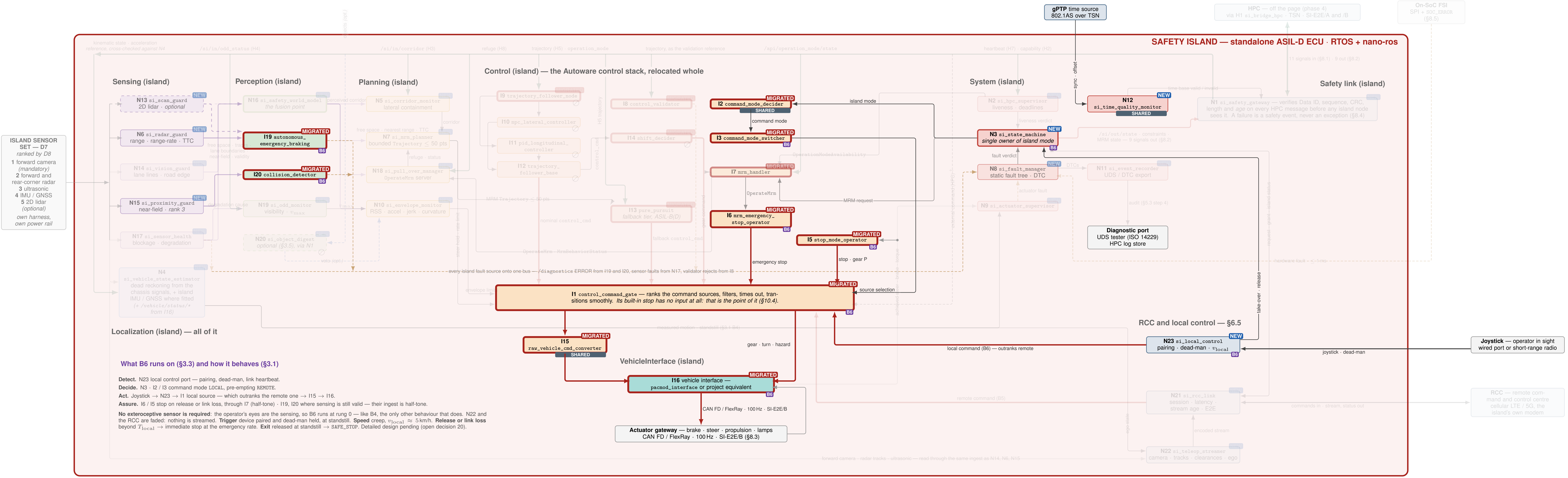
Phase 5 · B6 — Manual joystick control without streamed sensing

An operator in sight of the vehicle drives it with a joystick on the island's local control port — wired, or over a short-range radio link — with no sensor stream; the island holds the speed to a creep and stops on release or on loss of the link.
The phase-5 drawing, unchanged: what B6 runs on — and the SHARED nodes every behaviour runs through — is in full colour, what it merely passes through is half-tone, and the rest of the island is faded. *Detailed design pending (§6.5).*



gPTP time source
802.1AS over TSN
HPC — off the page (phase 4)
via H1 si_bridge_hpc · TSN · SI-E2E/A and /B
On-SoC FSI
SPI + SOC_ERROR
(§8.5)
objects (opt.)
SAFETY ISLAND — standalone ASIL-D ECU · RTOS + nano-ros
kinematic state · acceleration
reference, cross-checked against N4
/si/in/odd_status (H4)
/si/in/corridor (H3)
refuge (H8)
trajectory (H5) · operation_mode
trajectory, as the validation reference
/api/operation_mode/state
heartbeat (H7) · capability (H2)
11 signals in (§8.1) · 9 out (§8.2)
sync · offset
ISLAND SENSOR SET — D7
ranked by D8
1 forward camera (mandatory)
2 forward and rear-corner radar
3 ultrasonic
4 IMU / GNSS
5 2D lidar (optional)
own harness, own power rail
Sensing (island)
N13 si_scan_guard
2D lidar · optional
NEW
N6 si_radar_guard
range · range-rate · TTC
NEW
N14 si_vision_guard
lane lines · road edge
N15 si_proximity_guard
near-field · rank 3
NEW
N17 si_sensor_health
blockage · degradation
N4 si_vehicle_state_estimator
dead reckoning from the chassis signals, + island IMU / GNSS where fitted
(+ /vehicle/status/* from I16)
Localization (island) — all of it
Perception (island)
N16 si_safety_world_model
the fusion point
I19 autonomous_emergency_braking
MIGRATED
B6
I20 collision_detector
MIGRATED
B6
N19 si_odd_monitor
visibility · $v_{\max}$
N20 si_object_digest
optional (§3.5), via N1
free space · tracks · lane boundaries · near-field · validity
degradation cause
veto (opt.)
Planning (island)
N5 si_corridor_monitor
lateral containment
perceived corridor
corridor
free space · nearest range · TTC
N7 si_mrm_planner
bounded Trajectory ≤ 50 pts
refuge · status
N18 si_pull_over_manager
OperateMrm server
N10 si_envelope_monitor
RSS · accel · jerk · curvature
$v_{\max}$
Control (island) — the Autoware control stack, relocated whole
I9 trajectory_follower_node
I10 mpc_lateral_controller
I11 pid_longitudinal_controller
I12 trajectory_follower_base
I8 control_validator
H5 trajectory
I14 shift_decider
control_cmd
I13 pure_pursuit
fallback tier, ASIL-B(D)
MRM Trajectory ≤ 50 pts
nominal control_cmd
fallback control_cmd
gear command
OperateMrm · MrmBehaviorStatus
steer hold · rate limit
I2 command_mode_decider
MIGRATED
SHARED
island mode
command mode
I3 command_mode_switcher
MIGRATED
B6
OperationModeAvailability
I7 mrm_handler
MIGRATED
OperateMrm
MRM request
I6 mrm_emergency_stop_operator
MIGRATED
B6
emergency stop
I5 stop_mode_operator
MIGRATED
B6
stop · gear P
every island fault source onto one bus — /diagnostics ERROR from I19 and I20, sensor faults from N17, validator rejects from I8
envelope limits
I1 control_command_gate — ranks the command sources, filters, times out, transitions smoothly. Its built-in stop has no input at all: that is the point of it (§10.4).
MIGRATED
B6
source selection
external command (HPC) †
achieved steer · brake · torque
measured motion · standstill (§3.1 B4)
I15 raw_vehicle_cmd_converter
MIGRATED
SHARED
gear · turn · hazard
VehicleInterface (island)
I16 vehicle interface — pacmod_interface or project equivalent
MIGRATED
B6
CAN FD / FlexRay · 100 Hz · SI-E2E/B
Actuator gateway — brake · steer · propulsion · lamps
CAN FD / FlexRay · 100 Hz · SI-E2E/B (§8.3)
remote command (B5)
forward camera · radar tracks · ultrasonic — read through the same ingest as N14, N6, N15
System (island)
N2 si_hpc_supervisor
liveness · deadlines
liveness verdict
N3 si_state_machine
single owner of island mode
NEW
B6
fault verdict
N8 si_fault_manager
static fault tree · DTC
NEW
DTCs
actuator fault
N9 si_actuator_supervisor
N12 si_time_quality_monitor
NEW
SHARED
time base valid / invalid
N11 si_event_recorder
UDS / DTC export
audit (§5.3 step 4)
Diagnostic port
UDS tester (ISO 14229)
HPC log store
Safety link (island)
N1 si_safety_gateway — verifies Data ID, sequence, CRC, length and age on every HPC message before any island node sees it. A failure is a safety event, never an exception (§8.4)
/si/out/state · constraints · MRM state — 9 signals out (§8.2)
request · grant · island status
hardware fault · ≤ 1 ms
take-over · release
RCC and local control — §6.5
N23 si_local_control
pairing · dead-man · v_{local}
NEW
B6
local command (B6) — outranks remote
Joystick — operator in sight
wired port or short-range radio
joystick · dead-man
N21 si_rcc_link
session · latency · stream age · E2E
commands in · stream, status out
RCC — remote command and control centre
cellular LTE / 5G, the island's own modem
ego state
encoded stream
N22 si_teleop_streamer
camera · tracks · clearances · ego

What B6 runs on (§3.3) and how it behaves (§3.1)

Detect. N23 local control port — pairing, dead-man, link heartbeat.

Decide. N3 · I2 / I3 command mode `LOCAL`, pre-empting `REMOTE`.

Act. Joystick → N23 → I1 local source — which outranks the remote one → I15 → I16.

Assure. I6 / I5 stop on release or link loss, through I7 (half-tone) · I19, I20 where sensing is still valid — their ingest is half-tone.

No exteroceptive sensor is required: the operator's eyes are the sensing, so B6 runs at rung 0 — like B4, the only other behaviour that does. N22 and the RCC are faded: nothing is streamed. **Trigger** device paired and dead-man held, at standstill. **Speed** creep, $v_{\text{local}} \approx 5\,\text{km/h}$. **Release or link loss** beyond T_{local} → immediate stop at the emergency rate. **Exit** released at standstill → `SAFE_STOP`. Detailed design pending (open decision 20).

Stacks — the colours of `ArchitectureMain.pdf`

- Sensing
- Localization
- Perception
- Planning
- Control
- System
- VehicleInterface
- Vehicle
- Safety link

Where each node came from

- MIGRATED — the same Autoware node, migrated
- NEW — new — does not exist in Autoware

18 migrated + 19 new = **37**, plus the optional N20 — and N21–N23 of §6.5, outside the counts until their detailed design is done. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. † the HPC's external command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table; B5's remote command arrives through N21 instead.

Behaviours each node enables (§3.3)

one fixed slot per behaviour B6 — **B1** degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop · **B5** tele-operation · **B6** joystick. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

SHARED — **SHARED** — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.

Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.

⊘ **Used by no behaviour** — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

Messages

- → HPC-sourced, verified by N1
- → island sensor data
- → control command into the gate
- - - → fault · `/diagnostics`
- → actuator feedback
- → state · mode · decision
- - - → optional (§3.5)

In this view — B6, manual joystick control without streamed sensing

- **full colour** — named for B6 in §3.3 (Detect · Decide · Act · Assure), or SHARED (I2, I15, N12)
- **half-tone** — on B6's path by §3.1, but not named in §3.3
- **faded** — not used by B6. An edge is drawn at the weaker of its two ends. Only B6's own tag is drawn on each box.

Every edge on this page, with the design-document section it is taken from, is catalogued in `../messageflow/`, which draws the same 37 nodes coloured by execution partition instead.